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From data to value in smart waste management: Optimizing solid waste collection with a digital twin-based decision support system

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ABSTRACT

The importance of waste management, including collection, separation, recovery, and recycling, increases with the growing amount of waste. Technological innovations such as smart connected products, the Internet of Things, and digital twins are driving the development of smart management systems. Investments in necessary product-service systems are justified by cost savings and improved service quality, especially in affluent societies like Switzerland. However, there is a trade-off between cost savings and service quality that raises the question of optimal balance. Using a Swiss municipality as an example, this paper models the trade-off between cost savings and service quality using waste bin sensor modules. Simulation results demonstrate the impact of cost savings on service quality reduction and that substantial cost savings are possible without a service quality compromise. We also introduce a digital process twin as a decision support system that is able to leverage a growing database. These results contribute to research, firstly through the field study with 98 waste bins equipped with fill level sensor modules, secondly through the model-based analysis of the trade-off between cost savings and service quality, and thirdly by conceptualizing a digital twin-based decision support system. The results further contribute to practice, firstly by providing benchmarks for implementing similar systems in other municipalities without having to create their own simulations, secondly by presenting an innovative key performance indicator to measure service quality, and thirdly with a model that can be used for simulations to determine the individual optimum between costs and service quality.

1. Introduction

1.1. Relevance

Waste management is one of the three main objectives of the European Union's (EU) waste policy for protecting the environment and human health while promoting its transition to a circular economy [1]. Waste management deals with various activities, from waste collection and separation to waste recovery and recycling. It is a vital issue as the volume of waste is increasing rapidly due to steadily rising living standards and rapid urbanization [2] as well as industrial development and changing consumer behavior [3]. In 2018, the total waste generated by all households in the EU amounted to 698 million tons. The collection and management of this enormous quantity imposes huge costs, both economically and with respect to human resources, time, and environmental impact [1]. Switzerland is the focus of this study, as an innovation project from the municipality of Herzogenbuchsee in the Canton of Bern was the basis for the case study presented later in this paper. Switzerland is located in the heart of Europe, and its waste management challenges are comparable to other heavily populated regions of Western Europe. Considering such densely populated and highly developed economies such as Switzerland, it can be observed that the amount of waste per capita has stagnated for more than a decade after a phase of sizeable increase from 300 kilograms per year in 1970 to 700 kilograms per year in 2006, as seen in Fig. 1 [4]. This means that even in a small municipality like Herzogenbuchsee, with just under 10,000 inhabitants, more than 7000 tons of waste is generated annually. Due to the growing population in Switzerland, the absolute amount of waste continues to increase, which is why efficient waste management is becoming more relevant [5].

At the same time, large and dense residential areas and the urgent need for urban environmental protection create difficult conditions for waste management [6]. Therefore, to improve the quality of life for citizens and minimize the negative impact on the environment, efficient and proper waste management is a fundamental challenge [3].

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Fig. 1. Municipal waste per capita in Switzerland, 1970–2020 [4].

To achieve efficient waste management, stakeholders and operators must critically address the issues of cost and cost-effectiveness of infrastructure and processes.

1.2. Technologies and concepts

Information and communication technologies (ICT) offer a broad range of opportunities for developing solutions to the problems and challenges of waste management. Developments and progress in ICT transform traditional products into smart, connected products (SCP) by connecting them to the Internet of Things (IoT), which in turn transforms ecosystems and competition in many industries [7,8]. Therefore, the IoT is expected to change municipal waste management fundamentally in many areas, such as adaptive schedules for equipment use and maintenance, waste collection, and vehicle routing [9]. Furthermore, IoT and related concepts and technologies play a vital role in waste management systems as they allow for processing various types of information and thus help optimize the whole system [9]. SCP and smart systems allow for smart waste management (SWM) and are, therefore, considered an important part of smart city concepts [10]. The concept of the digital twin (DT), originated by Grieves [11], can be defined as a digital representation of a real-world counterpart, which can receive and provide data to create value within a use case [12]; despite this, there is still no common definition in the industry when it comes to the term DT [12,13]. With DTs, companies seek to create value in both the internal dimension (internal processes regarding product lifecycle) and the external dimension (during the usage phase in the market) [14–16]. The counterpart of a DT is often a physical object. However, in practice, any real-world entity with a recognizably distinct existence and relevance for creating value can be digitally represented by a DT [12]. In the case of processes, the term digital process twin (DPT) is used accordingly [17]. A real-world entity's representation in the digital realm can be categorized into two subsets. The first encompasses DTs symbolizing physical assets, frequently called "equipment twins". The second category, contrastingly, represents non-physical entities and is often referred to as "DPT". Such DPTs are more abstract, encapsulating processes, systems, or services instead of physical objects. They offer an innovative approach to visualizing, tracking, and enhancing various operational or business processes, thereby improving efficiency and productivity.

1.3. Research questions and innovative contribution

The focus of existing publications in the field of SWM concerns either maximizing profits, minimizing costs, or minimizing environmental impact through emissions by optimizing the routes of waste collection vehicles. The impact of SWM on service quality has been insufficiently explored until now, and the trade-off between cost savings and service quality has not yet been elaborated. It is, therefore, of great relevance to investigate the application of new technologies and concepts, such as DPT, in SWM with respect to their impact on cost savings and service quality. In this paper, we make an innovative contribution to ongoing research by addressing the following research questions:

- RQ 1 How can decision-makers balance the trade-off between cost savings and service quality in waste management?
- RQ 2 How can a DPT, in combination with an area-wide sensor module system, create value for decision-makers in waste management companies?
- RQ 3 How can a DPT be designed, and what specific functions are essential for a DPT to create value?

1.4. Structure of the paper

The remainder of this paper is structured as follows: First, we present a literature review (Section 2), elaborating on the key research and findings regarding SWM and the process by which we derived the research gaps and composed the research questions. Section 3 presents the methods and the research procedure used to answer the research questions, while Section 4 introduces and explains the use case for this study. In Section 5, the results are presented along with the main topics of data resources, the generated SWM model, and achieved optimizations. Section 6 then discusses the major findings along with